

Heat Recovery Ventilators and Whole House Fans

Bringing fresh air in without throwing the heat away — and the one fan that can be genuinely dangerous in winter.

Heat recovery ventilator (HRV)

The device that transfers heat from outgoing air to incoming fresh air. Two airstreams pass close without mixing, so the outgoing air warms the incoming.

ERV

The same idea, but it transfers moisture as well as heat.

YOU HAVE MISSED THIS

What recovery mechanical ventilation system requires a condensate disposal system?

HRV.

Warm moist outgoing air meets a cold surface inside the core and condenses there, so the unit produces water and needs somewhere to put it. An ERV moves that moisture across to the incoming air instead, so it produces much less condensate.

The numbers here

HRV or ERV exterior opening — minimum distance from a door, operable window or property line

3 ft

Three feet stops the exhaust being drawn straight back into the intake or into a neighboring opening.

The two duct configurations the exam names

YOU HAVE MISSED THIS 3 TIMES

On an HRV system, the _____ duct configuration relies on the local exhaust fans to remove most pollutants and on the HVAC system fan to circulate outside ventilation air.

Simplified.

YOU HAVE MISSED THIS 2 TIMES

What type of ducted HRV or ERV configuration uses exhaust ducts connected to grilles in bathrooms, kitchens and utility rooms to remove pollutants and provide outside ventilation air to the living areas through the HVAC return duct?

Partially ducted.

Read the two by how much dedicated ductwork the HRV has of its own. Simplified has none — it leans entirely on the existing exhaust fans and the HVAC fan. Partially ducted has its own exhaust grilles in the wet rooms but still borrows the HVAC return to distribute the fresh air. Fully ducted, the third case, has both its own supply and its own exhaust.

Whole house fans

Whole house fan

Draws air from the house and discharges it into the attic. In winter it lets large amounts of conditioned air escape into the attic.

YOU HAVE MISSED THIS 2 TIMES

What type of fan should NOT be used in the winter when a combustion appliance draws combustion air from inside the house?

Whole house fan.

This is a safety answer, not an energy one. A whole house fan pulls a large volume out of the building and drops the indoor pressure. An atmospherically vented furnace or water heater relies on its exhaust rising up the flue, and under negative pressure that flow can reverse — pulling combustion products, including carbon monoxide, back into the house.

Say it back

Cover the page. Answer each one out loud.

1. What does an HRV do, and how does an ERV differ?
2. Which of the two needs a condensate disposal system, and why?
3. How far must an HRV exterior opening sit from a door or window?
4. Describe the simplified configuration.
5. Describe the partially ducted configuration.
6. What does a whole house fan do in winter, and why is it dangerous with a combustion appliance?